

JEE MAIN REVISION MASTER: HALOALKANES AND HALOARENES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 177 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	SN1 and SN2 Nucleophilic Substitution	143 PYQs	Reaction Mechanisms
Very High	Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	36 PYQs	Elimination & Aromatic

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
SN1 and SN2 Nucleophilic Substitution	$\text{SN1: Rate} \propto [\text{RX}]$; $(3^\circ > 2^\circ > 1^\circ)$, Racemization , $\text{SN2: Rate} \propto [\text{RX}][\text{Nu}]$; $(1^\circ > 2^\circ > 3^\circ)$, Inversion
Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	$\text{Saytzeff Rule: Highly substituted, more stable alkene is the major product in dehydrohalogenation}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 SN1 and SN2 Nucleophilic Substitution (Very High Yield — 143 PYQs)

Core Principle: Kinetics, stereochemical outcomes, carbocation rearrangements in SN1, and concerted backside displacement in SN2.

■ **Caution / JEE Pitfall:** Polar PROTIC solvents (e.g. H_2O , ROH) stabilize carbocations and favor SN1. Polar APROTIC solvents (e.g. acetone, DMSO) favor SN2 by desolvating nucleophiles.

■ **Speed Trick:** Allylic and benzylic halides undergo BOTH SN1 (resonance-stabilized carbocation) and SN2 (low steric hindrance) at remarkably high rates.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Due to increase in van der Waals forces, the order of boiling points is $\text{CH}_3\text{CH}_2\text{CH}_2\text{I} > \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Given below are two statements : Statement (I) : Benzyl chloride reacts faster in $\text{S}_{\text{N}}1$ mechanism than ethyl chloride. Statement (II) : $\text{E} > \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider the following compounds Arrange these compounds in the increasing order of reactivity with nitrating mixture..." [Ans: N/A]

3.2 Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions (Very High Yield — 36 PYQs)

Core Principle: β -elimination mechanism, Saytzeff vs Hofmann regioselectivity, and unreactivity of haloarenes toward nucleophilic substitution.

■ **Caution / JEE Pitfall:** Haloarenes (like chlorobenzene) are extremely UNREACTIVE toward nucleophilic substitution due to partial double bond character (C-Cl resonance) and sp^2 carbon.

■ **Speed Trick:** Presence of electron-withdrawing groups (like $-NO_2$) at ortho- and para-positions dramatically increases reactivity of haloarenes toward nucleophilic substitution.

Representative Source PYQs:

• **[2026 Morning Shift] Q#3 (MCQ):** "Match the LIST-I with LIST-II List-I Name of reaction $\begin{aligned} \text{List-...} \end{aligned}$ "
[Ans: B]

• **[2026 Morning Shift] Q#3 (MCQ):** "The correct order of the rate of reaction of the following reactants with nucleophile by S_N1 mechanism is : (Given : Structures I and I...)" [Ans: N/A]

• **[2026 Evening Shift] Q#4 (MCQ):** "Correct statements regarding alkyl halides ($R-X$) among the following are : A. Alcohol being less polar solvent as compared to water, alcohol..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

The correct order of the rate of the reaction for the following reaction with respect to nucleophiles is : $CH_3Br + Nu^- \rightarrow CH_3Nu + Br^-$

- (A) $ClO_4^- \rightarrow CH_3COO^- \rightarrow OH^- \rightarrow P^{\ominus}O^-$
- (B) $OH^- \rightarrow P^{\ominus}O^- \rightarrow CH_3COO^- \rightarrow ClO_4^-$ [Correct]
- (C) $CH_3COO^- \rightarrow P^{\ominus}O^- \rightarrow OH^- \rightarrow ClO_4^-$
- (D) $P^{\ominus}O^- \rightarrow OH^- \rightarrow CH_3COO^- \rightarrow ClO_4^-$

Correct Answer: B

Solution Analysis: For CH_3Br (a methyl halide), the reaction proceeds by S_N2 , so the rate increases with nucleophilicity of Nu^- . For oxygen nucleophiles in the same row, nucleophilicity roughly follows basicity, and any resonance stabilization of the anion reduces nucleophilicity: OH^- : strong base, charge localized $\Rightarrow$ best nucleophile
 $P^{\ominus}O^-$: resonance-stabilized $\Rightarrow$ less nucleophilic than OH^-
 CH_3COO^- : even more resonance delocalization over two oxygens $\Rightarrow$ weaker nucleophile
 ClO_4^- : highly resonance-stabilized, essentially non-nucl

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Morning Shift (Q#3)

Match the LIST-I with LIST-II List-I Name of reaction List-II Reagent or catalyst used A. Finkelstein reaction I. SbF_3 B. Swarts reaction II. Na, dry ether C. Sandmeyer's reaction III. NaI D. Fittig reaction IV. Cu_2Cl_2 Choose the correct answer from the options given below :

- (A) -I, B-IV, C-III, D-III
- (B) -III, B-I, C-IV, D-II **[Correct]**
- (C) -IV, B-II, C-I, D-III
- (D) -I, B-III, C-II D-IV

Correct Answer: B

Solution Analysis: A. Finkelstein reaction: This is a halogen exchange reaction used specifically for the preparation of alkyl iodides. In this reaction, alkyl chlorides or bromides are treated with sodium iodide (NaI) in dry acetone. Therefore, A matches with III. B. Swarts reaction: This reaction is the preferred method for the synthesis of alkyl fluorides. It involves heating an alkyl chloride or bromide in the presence of a metallic fluoride. Some of the common fluorinating agents used are AgF , Hg_2F_2 , CoF_2 , or antimony trifluoride (SbF_3). Therefore, B matches with I. C. Sandmeyer